Deep Hyperspectral and Multispectral Image Fusion via Probabilistic Matrix Factorization

Baihong Lin and Yulan Guo, *Senior Member, IEEE*

***Abstract*— Deep learning methods are popular for hyperspectral and multispectral image (HSI-MSI) fusion to obtain a high-resolution HSI. However, most of them are unsatisfactory due to limited generalization ability and poor interpretability. This article proposes a highly interpretable deep HSI-MSI fusion method based on probabilistic matrix factorization (PMF) under the Bayesian framework. In the proposed method, an HSI is factorized into two matrices, namely, the Gaussian-prior-regularized spectral matrix and the deep-prior-regularized abundance matrix. Then, we split the optimization process into two meaningful iterative updating steps: updating the spectral matrix based on least-squares estimation and updating the abundance matrix based on a convolutional neural network (CNN)-based Gaussian denoiser for 2-D gray images. To improve the generalization ability, we provide solutions for selections of hyperparameters, CNN-based denoiser architecture, and training strategy. Using the given solutions, the proposed fusion method can be trained with 2-D images once and then used to fuse different types of HSI and MSI excellently. Experiments on three datasets demonstrate that the proposed fusion method has good fusion performance and high generalization ability compared with other state-of-the-art methods. The source code will be available at https://github.com/KevinBHLin/.**

***Index Terms*— Convolutional neural network (CNN)-based Gaussian denoiser, deep prior regularization (DPR), hyperspectral image (HSI) fusion, inertial Proximal Alternating Linearized Minimization (iPALM), probabilistic matrix factorization (PMF).**

I. Introduction

HYPERSPECTRAL image (HSI) has been widely used in many areas, e.g., remote sensing [1] and medical applications [2]. This kind of image has hundreds of spectral bands. Unfortunately, it usually features low spatial resolution due to the limitation of the imaging instrument [3]. To enhance the spatial resolution of an HSI and meet the requirement of further image applications, several methods [4], [5], [6] are proposed to fuse an HSI with a multispectral image (MSI) or RGB image, which features low spectral resolution and high spatial resolution. Since then, the design of an efficient HSI-MSI fusion method turns to be an interesting research topic.

Numerous HSI-MSI fusion methods have been proposed in the literature. Early studies focus on the component substitution method [4]. This kind of fusion method usually obtains high-resolution HSI by substituting the components of transformed HSI with the corresponding high spatial resolution MSI or RGB image, e.g., the principal component analysis (PCA) method. This fusion method does not have a strict mathematical framework, and its results are usually poor with high spectral distortions [4]. Later, various fusion models have been introduced with the development of mathematical techniques. More recently, popular HSI-MSI fusion methods can be roughly divided into two categories, including the Bayesian methods and the deep learning methods.

A Bayesian method models the HSI-MSI fusion task as an ill-posed mathematical problem and turns it into a maximum a posteriori (MAP) or expectation–maximization (EM) optimization problem constrained by various HSI priors. It is obvious that prior modeling of the fusion HSI is the bottleneck in this kind of fusion method. So far, various designs of the image prior have been proposed, including total-variation-based priors [6], tensor-decomposition-based priors [7], [8], sparsity-based priors [8], [9], low rank priors [10], [11], and nonnegativity priors [12]. Based on these prior designs, the Bayesian method can jointly make full use of known imaging conditions and data priors to achieve good fusion performance with high interpretability and generalization ability. However, due to the complex forms of prior functions, the Bayesian method usually introduces a complicated optimization process. Furthermore, this method requires accurate and manual tuning of hyperparameters based on a lot of experiments; otherwise, it would perform poorly with high spectral distortions. Besides, it is unknown whether these designed priors with an explicit form can accurately represent the image data distribution.

The deep learning method models the HSI-MSI fusion task as a regression problem based on the deep convolutional neural network (CNN). Different from the Bayesian method that requires various prior function designs, the deep learning method focuses on the generation of a good HSI training dataset and the design of an efficient fusion network to learn data prior from the training process. Currently, there have been several HSI datasets, e.g., CAVE [13], and BGU ICVL [14]. Based on these datasets, various fusion neural networks have been designed, e.g., autoencoder (AE)-based networks [15] and generative-adversarial-network (GAN)-based

Manuscript received 17 October 2022; revised 12 January 2023 and 9 February 2023; accepted 12 February 2023. Date of publication 14 February 2023; date of current version 23 February 2023. This work was supported in part by the Natural Science Foundation of Guangdong Province (China) under Grant 2022A1515010493, in part by the Fundamental Research Funds for the Central Universities (Sun Yat-sen University) under Grant 2021qntd11, in part by the Science Technology and Innovation Commission of Shenzhen Municipality under Grant 2021A03, and in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2022B1515020103. *(Corresponding author: Baihong Lin.)*

The authors are with the School of Electronics and Communication Engineering, Sun Yat-sen University, Shenzhen 518107, China (e-mail: linbh6@mail.sysu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3244992

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

networks [16], [17]. These networks achieve promising fusion results since deep learning models can effectively extract features from training datasets. However, most deep-learning-based fusion methods have several limitations compared with the Bayesian method.

1) *Limited Generalization Ability:* Most deep learning solutions highly rely on training datasets to achieve good fusion performance. They cannot work well on the test datasets with spectral bands different from training datasets.
2) *Poor Interpretability:* Most fusion networks are designed based on universal deep learning models, which can also be adopted for other image processing tasks. They usually lack interpretability. Besides, known prior information cannot be utilized directly and flexibly, e.g., some known priors of an HSI and the imaging relationships between HSI and MSI.

It is obvious that the Bayesian method and the deep learning method have drawbacks for HSI-MSI fusion, but they offer complementary advantages. Thus, to obtain a fusion model that can utilize known priors for inference and extract features from training data jointly, these two types of fusion methods have been combined in recent years. So far, there exits two schemes to combine the Bayesian method and the deep learning method. The first scheme designs a specific deep neural fusion network based on the unfolded optimization process of a Bayesian model to improve interpretability, e.g., DBI-Net [18], USD-Net [19], and others [20], [21]. Since the known imaging relationships between HSI and MSI can be utilized in the design of network architecture, this scheme usually produces excellent fusion performance. Unfortunately, most known designed HSI data priors cannot be directly utilized in this scheme. Besides, the most designed network still requires extra training datasets; thus, it cannot overcome the drawbacks of limited generalization ability.

The second scheme designs a Bayesian model using deep CNN, e.g., CNN-FUS [22], NVPGM [23], MIAE [24], and others [25]. In this scheme, the Bayesian model contains a mathematically implicit data-driven prior function constructed based on CNN. The CNN-based prior function is also called deep prior or plug-and-play prior when it has a mathematically implicit form and can be split into a subproblem solved by a deep CNN-based denoiser [12], [22]. Since this scheme is based on the Bayesian framework, different known HSI priors can be utilized flexibly, and the designed model has high interpretability. Besides, the CNN adopted in this scheme can be trained using gray images; thus, it does not require an extra HSI training dataset, i.e., it is a kind of unsupervised deep learning method with high generalization ability. However, its fusion performance depends on the prior modeling, the splitting optimization methods, and the designed neural network. So far, it is challenging to design a good and robust deep-prior-based fusion model.

In this article, we propose a novel deep HSI-MSI fusion method based on probabilistic matrix factorization (PMF) under the Bayesian framework. Specifically, the fusion image is factorized into two matrices, namely, the spectral matrix and the abundance matrix. Then, the spectral matrix is regularized by Gaussian prior to restrict the corresponding values, whereas the abundance matrix is regularized by a mathematically implicit deep prior function to reconstruct the spatial texture. To efficiently obtain a good fusion result, we optimize the proposed model using the block coordinate descent method [26] and the inertial Proximal Alternating Linearized Minimization (iPALM) method [27], dividing the optimization process into two highly interpretable iterative parts: the spectral matrix updating step based on least-squares estimation and the abundance matrix updating step using a CNN-based gray image denoiser for Gaussian noise. Experiments on three datasets demonstrate the good fusion performance of the proposed fusion model. The main contributions of the work are summarized as follows.

1) We propose a highly interpretable deep HSI-MSI fusion method based on a 2-D image CNN-based Gaussian denoiser, which can fuse the HSI and the MSI under a known imaging system and achieve good fusion performance.
2) We introduce the block coordinate descent method and the iPALM method to efficiently solve the proposed nonconvex fusion framework, which can help to accelerate the convergence to reach a critical point.
3) We provide solutions for selections of hyperparameters, CNN-based denoiser architecture, and training strategy so that the proposed method can be trained with 2-D images once and then used to excellently fuse different types of HSI and MSI.

The remainder of this article is organized as follows. Section II formulates the fusion problem from a probabilistic perspective. Section III introduces the proposed deep fusion method based on PMF. Section IV introduces the solutions for selections of hyperparameters, CNN-based denoiser architecture, and training strategy, and compares the proposed method with other fusion methods based on experiments. The conclusion is drawn in Section V.

II. Problem Formulation

Let $\mathbf{Z} \in \mathbb{R}^{L\times MN}$ denote a $M \times N \times L$ high spatial resolution HSI; let $\mathbf{Y}_h \in \mathbb{R}^{L\times mn}$ and $\mathbf{Y}_m \in \mathbb{R}^{l\times MN}$ denote the corresponding $m \times n \times L$ low spatial resolution HSI and $M \times N \times l$ MSI or RGB image corrupted by Gaussian noises $\mathbf{N}_h$ and $\mathbf{N}_m$, respectively. L and l denote the numbers of spectral channel bands. Then, the relationship among these images $\mathbf{Z}$, $\mathbf{Y}_h$, and $\mathbf{Y}_m$ can be formulated as

$$\mathbf{Y}_h = \mathbf{ZBS} + \mathbf{N}_h \tag{1}$$

$$\mathbf{Y}_m = \mathbf{RZ} + \mathbf{N}_m \tag{2}$$

where $\mathbf{R}$ denotes the spectral response matrix, $\mathbf{B}$ denotes the blurring matrix of the hyperspectral imaging system, and $\mathbf{S}$ denotes the downsampling matrix.

In our problem, we assume that the parameters related to the imaging system in $\Psi = \{\mathbf{R}, \mathbf{B}, \mathbf{S}\}$ are all known, and the elements of $\mathbf{N}_m$ and $\mathbf{N}_h$ are drawn from independent zero-mean Gaussian distributions. Since the noises from different spectral bands are usually different, we assume that $(\mathbf{N}_m)_{ij} \sim \mathcal{G}(0, \sigma_{m_i}^2)$ and $(\mathbf{N}_h)_{ij} \sim \mathcal{G}(0, \sigma_{h_i}^2)$. Then,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

using (1)–(5), the distributions of $\mathbf{Y}_m$ and $\mathbf{Y}_h$ given $\mathbf{Z}$ can be formulated as

$$\ln p(\mathbf{Y}_h|\mathbf{Z}, \Psi) \propto -\frac{1}{2}\|\mathbf{Y}_h - \mathbf{ZBS}\|_{\Sigma_h}^2 \tag{3}$$

$$\ln p(\mathbf{Y}_m|\mathbf{Z}, \Psi) \propto -\frac{1}{2}\|\mathbf{Y}_m - \mathbf{RZ}\|_{\Sigma_m}^2 \tag{4}$$

where $\|\mathbf{X}\|_\Sigma^2 = \mathrm{tr}[\mathbf{X}^\mathrm{T}\Sigma^{-1}\mathbf{X}]$, and $\mathrm{tr}[\cdot]$ denotes the matrix trace. Σ_h and Σ_m are diagonal covariance matrices, $\Sigma_h = \mathrm{diag}(\sigma_{h1}^2, \sigma_{h2}^2, \ldots, \sigma_{hL}^2)$, and $\Sigma_m = \mathrm{diag}(\sigma_{m1}^2, \sigma_{m2}^2, \ldots, \sigma_{mL}^2)$. This article aims at obtaining the image $\mathbf{Z}$ given the image $\mathbf{Y}_h$, the image $\mathbf{Y}_m$, the covariance matrices Σ_h and Σ_m, and the parameters of the imaging system $\Psi = \{\mathbf{R}, \mathbf{B}, \mathbf{S}\}$. It is obvious that the above problem is ill-posed. To solve the problem, more information about the image $\mathbf{Z}$ is required. In Section III, we will discuss the known property of image $\mathbf{Z}$ and model the corresponding prior.

III. Proposed HSI-MSI Fusion Method

A. Probabilistic Matrix Factorization Fusion Framework

According to the linear spectral mixing model (LSMM) [28], a pixel of the HSI can be regarded as a linear combination of several spectral endmembers. Assume that the image $\mathbf{Z}$ contains of r endmembers, $r \ll L$, each of which can be denoted as $\mathbf{u}_j \in \mathbb{R}^{L\times 1}$, $j = 1, 2, \ldots, r$. Then, the image $\mathbf{Z}$ can be represented as

$$\mathbf{Z} = \mathbf{UV}, \quad \mathbf{U}_{i,j} \geq 0, \quad \mathbf{V}_{i,j} \geq 0 \quad \forall i, j, \quad \mathbf{1}^\mathrm{T}\mathbf{V} = \mathbf{1}^\mathrm{T} \tag{5}$$

where $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_r]$ denotes the spectral matrix and $\mathbf{V}$ denotes the abundance matrix, in which $\mathbf{V}_{i,j}$ represents the proportion of the ith spectral endmember contained in the jth pixel. This model is very popular for different types of HSI processing tasks. However, although LSMM is well known in HSI processing, such a model usually requires complicated optimization for HSI-MSI fusion due to the nonnegativity constraint and the sum-to-one constraint [29], [30], [31].

To simplify the optimization, we study the problem from a probabilistic perspective and build the fusion framework based on PMF [32]. According to our previous work based on the PMF [5], the image $\mathbf{Z}$ can be represented as

$$\mathbf{Z} = \mathbf{HUV} \tag{6}$$

where $\mathbf{H} \in \mathbb{R}^{L\times d}$ is a known spectral subspace matrix, which can be obtained from the low spatial resolution HSI $\mathbf{Y}_h$ using truncated SVD or VCA [33]. $\mathbf{HU}$ and $\mathbf{V}$ denote latent spectral and texture feature matrices, respectively, and elements of $\mathbf{U} \in \mathbb{R}^{d\times r}$ and $\mathbf{V} \in \mathbb{R}^{r\times MN}$ are assumed to be drawn from two zero-mean Gaussian distributions, i.e., $\mathbf{U}_{i,j} \sim \mathcal{N}(0, \lambda_u^{-1})$ and $\mathbf{V}_{i,j} \sim \mathcal{N}(0, \lambda_v^{-1})$. The above PMF-based model has been demonstrated successful for HSI-MSI fusion in [5]. Unfortunately, texture information laid in each row of $\mathbf{V}$ is not considered in [5] since elements of $\mathbf{V}$ are only regularized by Gaussian priors.

In our modeling, instead of adopting the Gaussian prior assumption, we assume that the prior of $\mathbf{V}$ has the following form:

$$\ln p(\mathbf{V}) \propto \lambda_v \mathcal{T}(\mathbf{V}) = \lambda_v \sum_{i=1}^{r} \mathcal{T}(\mathbf{V}_{i.}) \tag{7}$$

where $\mathbf{V}_{i.}$ denotes the ith row of $\mathbf{V}$, λ_v is a hyperparameter, and $\mathcal{T}(\cdot)$ is a texture regularization function. This assumption is reasonable since texture information or spatial relation laid in each row of the abundance matrix $\mathbf{V}$ should be fully considered to enhance fusion performance. In previous studies, various explicit texture regularization functions have been designed, e.g., tensor-decomposition-based regularization [7]. However, we do not adopt these explicit regularization functions since they usually feature limited texture modeling abilities. Recently, deep prior learning attracts increasing interests [22], [34], [35] since this kind of model can learn prior from data comprehensively and efficiently. Inspired by the concept of deep prior learning, we assume that $\mathcal{T}(\cdot)$ is a deep prior regularization (DPR) function, i.e., the following problem about $\mathcal{T}(\cdot)$ can be solved by a CNN-based Gaussian denoiser:

$$\begin{aligned}\mathbf{V}_{i.}^{\mathrm{opt}} &= \arg\min_{\mathbf{V}_{i.}} \frac{1}{2\sigma^2}\|\mathbf{V}_{i.} - \mathbf{x}\|_2^2 + \mathcal{T}(\mathbf{V}_{i.}) \\ &= \mathrm{CNNDenoiser}\left(\mathbf{x}, \sigma^2\right).\end{aligned} \tag{8}$$

Based on the above modeling, the fusion problem is equivalent to the estimation of $\mathbf{U}$ and $\mathbf{V}$. According to the Bayesian theorem, the posterior of $\mathbf{U}$ and $\mathbf{V}$ satisfies

$$\begin{aligned}&p(\mathbf{U}, \mathbf{V}|\mathbf{Y}_h, \mathbf{Y}_m, \Psi, \Sigma_h, \Sigma_m, \lambda_u, \lambda_v) \\ &\quad \propto p(\mathbf{Y}_h|\mathbf{U}, \mathbf{V}, \Psi)p(\mathbf{Y}_m|\mathbf{U}, \mathbf{V}, \Psi)p(\mathbf{U})p(\mathbf{V}).\end{aligned} \tag{9}$$

Then, based on MAP estimation, we can obtain the following optimization problem:

$$\arg\min_{\mathbf{U},\mathbf{V}} \frac{1}{2}\Phi(\mathbf{U}, \mathbf{V}) + \frac{1}{2}\lambda_u\|\mathbf{U}\|_2^2 + \lambda_v\mathcal{T}(\mathbf{V}) \tag{10}$$

$$\Phi(\mathbf{U}, \mathbf{V}) = \|\mathbf{Y}_h - \mathbf{HUVBS}\|_{\Sigma_h}^2 + \|\mathbf{Y}_m - \mathbf{RHUV}\|_{\Sigma_m}^2. \tag{11}$$

B. Block Coordinate Descent Optimization Procedure

The optimization problem (10) is nonconvex since $\mathcal{T}(\cdot)$ is a highly nonconvex function. However, since $\Phi(\mathbf{U}, \mathbf{V})$ is a multiconvex function, we try to optimize the problem defined in (10) using the block coordinate descent method [26], in which the variables are updated as follows:

$$\mathbf{V}^{(k+1)} \in \arg\min_{\mathbf{V}} \frac{1}{2}\Phi\left(\mathbf{U}^{(k)}, \mathbf{V}\right) + \lambda_u\mathcal{T}(\mathbf{V}) \tag{12}$$

$$\mathbf{U}^{(k+1)} \in \arg\min_{\mathbf{U}} \frac{1}{2}\Phi\left(\mathbf{U}, \mathbf{V}^{(k+1)}\right) + \frac{1}{2}\lambda_v\|\mathbf{U}\|_2^2. \tag{13}$$

The above method divides the optimization procedure into two subproblems: *Updating the Matrix* $\mathbf{V}$ and *Updating the Matrix* $\mathbf{U}$.

1) Updating the Matrix $\mathbf{V}$*:* It is obvious that we cannot directly solve the problem (12) since the function $\mathcal{T}(\cdot)$ does not have an explicit form. As $\mathcal{T}(\cdot)$ represents the DPR, we have to relate the problem (12) to the problem (8). Inspired by Lanaras et al. [29] and Lin et al. [34], we introduce the proximal forward–backward scheme to solve the problem (12). Since the iPALM method can accelerate the convergence of the proximal forward–backward scheme [27], we optimize the problem (12) as the following form based on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

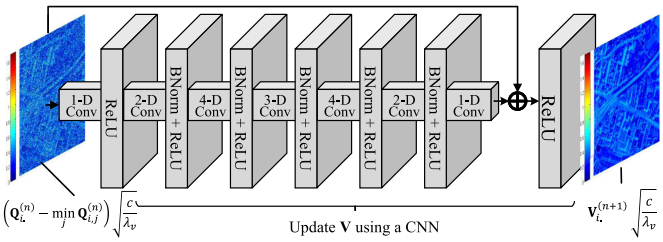


Fig. 1. DnCNN Gaussian denoiser for (21). More details about the network can be found in [34].

iPALM method:

$$\tilde{\mathbf{V}}^{(n)} = \mathbf{V}^{(k,n)} + \frac{n-1}{n+2}\left(\mathbf{V}^{(k,n)} - \mathbf{V}^{(k,n-1)}\right) \tag{14}$$

$$\mathbf{Q}^{(n)} = \tilde{\mathbf{V}}^{(n)} - \frac{1}{2c}\nabla_{\mathbf{V}}\Phi\left(\mathbf{U}^{(k)}, \tilde{\mathbf{V}}^{(n)}\right) \tag{15}$$

$$\begin{aligned}\nabla_{\mathbf{V}}\Phi(\mathbf{U},\mathbf{V}) &= 2\mathbf{U}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}}\Sigma_h^{-1}(\mathbf{HUVBS} - \mathbf{Y}_h)\mathbf{S}^{\mathrm{T}}\mathbf{B}^{\mathrm{T}} \\ &\quad + 2\mathbf{U}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{\mathrm{T}}\Sigma_m^{-1}(\mathbf{RHUV} - \mathbf{Y}_m)\end{aligned} \tag{16}$$

$$\mathbf{V}^{(k,n+1)} = \arg\min_{\mathbf{V}} \frac{c}{2\lambda_v}\left\|\mathbf{V} - \mathbf{Q}^{(n)}\right\|_2^2 + \mathcal{T}(\mathbf{V}) \tag{17}$$

where $c = \gamma\|\mathbf{U}^{\mathrm{T}}\mathbf{H}^{\mathrm{T}}\mathbf{R}^{\mathrm{T}}\Sigma_m^{-1}\mathbf{RHU}\|_F$ is a positive constant based on the Lipschitz continuity [27], and $\gamma > 1$ is a fixed parameter. Based on the above optimization scheme, the problem (12) can be solved by two iterative updating steps, as shown in (15) and (17). According to (7) and (8), (17) is equivalent to the following:

$$\mathbf{V}_{i.}^{(n+1)} = \mathrm{CNNDenoiser}\left(\mathbf{Q}_{i.}^{(n)}, \frac{\lambda_v}{c}\right) \tag{18}$$

where $\mathbf{V}_{i.}^{(n+1)}$ and $\mathbf{Q}_{i.}^{(n)}$ denote the ith rows of $\mathbf{V}^{(n+1)}$ and $\mathbf{Q}^{(n)}$, respectively, and (λ_v/c) denotes the variance of the Gaussian noise. The design of the CNN-based denoiser turns out to be a key problem of the proposed fusion method (see Section III-C).

Note that the above process of updating the abundance matrix $\mathbf{V}$ is highly interpretable. Specifically, (15) updates the matrix $\mathbf{V}$ using the gradient descend method based on the spectral matrix $\mathbf{HU}$, whereas (18) corrects the updating results of (15) by taking errors as Gaussian noises and utilizing the CNN-based denoiser to remove them. Given the spectral matrix $\mathbf{HU}$, these two iterative updating steps can obtain an optimized abundance matrix $\mathbf{V}$.

2) Updating the Matrix $\mathbf{U}$*:* Equation (13) is a least-squares problem since the variable $\mathbf{V}$ is fixed. Thus, we force the derivative of (13) with respect to $\mathbf{U}$ to be 0 and can obtain the following solution:

$$\begin{aligned}\Sigma_u &= \mathbf{W}^{(k+1)}\left(\mathbf{W}^{(k+1)}\right)^{\mathrm{T}} \otimes \mathbf{H}^{\mathrm{T}}\Sigma_h^{-1}\mathbf{H} \\ &\quad + \mathbf{V}^{(k+1)}\left(\mathbf{V}^{(k+1)}\right)^{\mathrm{T}} \otimes \mathbf{H}^{\mathrm{T}}\mathbf{R}^{\mathrm{T}}\Sigma_m^{-1}\mathbf{RH} + \lambda_u\mathbf{I}_{dr}\end{aligned} \tag{19}$$

$$\begin{aligned}\mathrm{vec}\left[\mathbf{U}^{(k+1)}\right] &= \Sigma_u^{-1}\mathrm{vec}\left[\mathbf{H}^T\Sigma_h^{-1}\mathbf{Y}_h\left(\mathbf{W}^{(k+1)}\right)^{\mathrm{T}}\right. \\ &\quad \left. + \mathbf{H}^T\mathbf{R}^T\Sigma_m^{-1}\mathbf{Y}_m\left(\mathbf{V}^{(k+1)}\right)^{\mathrm{T}}\right]\end{aligned} \tag{20}$$

Algorithm 1 Proposed DPMF Fusion Method

Input:
1) $\mathbf{Y}_h \in \mathbb{R}^{L\times mn}$ denotes the low spatial resolution HSI
2) $\mathbf{Y}_m \in \mathbb{R}^{l\times MN}$ denotes the high spatial resolution MSI
3) Σ_h and Σ_m denote the noise covariance matrices
4) The parameters of the imaging system, $\mathbf{R}$, $\mathbf{B}$, and $\mathbf{S}$
5) The hyperparameters λ_u, λ_v, d and r (the size of $\mathbf{U}$)
6) The CNN-based Gaussian denoiser
7) The iteration numbers T_{in} and T_o

Initialize:
1) $k = 0$
2) Obtain $\widehat{\mathbf{H}} \in \mathbb{R}^{L\times\max(d,r)}$ from $\mathbf{Y}_h$ using SVD
3) Let $\mathbf{H}$ be the first d columns of the matrix $\widehat{\mathbf{H}}$
4) Initialize $\mathbf{U}_{i,j}^{(0)} = 1$ when $i = j$, otherwise, $\mathbf{U}_{i,j}^{(0)} = 0$
5) Initialize $\mathbf{W}^{(0)} = \mathbf{H}^T\mathbf{Y}_h$ and $\mathbf{V}^{(0)} = \widehat{\mathbf{H}}^T\widetilde{\mathbf{Y}}_h$, where $\widetilde{\mathbf{Y}}_h \in \mathbb{R}^{L\times MN}$ denotes the interpolation image of $\mathbf{Y}_h$.

Do:
1) $k \leftarrow k + 1$
2) $\mathbf{V}^{(k,n=1)} \leftarrow \mathbf{V}^{(k,n=0)} \leftarrow \mathbf{V}^{(k-1)}$
3) **For** $n \leftarrow 1$ to T_{in}
   a) Update $\tilde{\mathbf{V}}^{(n)}$ using Eqn.(14).
   b) Update $\mathbf{Q}^{(n)}$ using Eqn.(15) and Eqn.(16).
   c) Update $\mathbf{V}^{(k,n)}$ using Eqn.(18).
4) Update $\mathbf{V}^{(k)}$: $\mathbf{V}^{(k)} \leftarrow \mathbf{V}^{(k,n=T_{in})}$
5) Update $\mathbf{U}^{(k)}$ using Eqn.(19) and Eqn.(20).

While $|\frac{\Phi(\mathbf{U}^{(k-1)},\mathbf{V}^{(k-1)})}{\Phi(\mathbf{U}^{(k)},\mathbf{V}^{(k)})} - 1| < \varepsilon$ **and** $k < T_o$.

Output: The fusion image $\mathbf{Z} = \mathbf{HU}^{(T_o)}\mathbf{V}^{(T_o)}$.

where vec[·] is the vec-operator [36], $\otimes$ represents the Kronecker product, $\mathbf{W}^{(k+1)} = \mathbf{V}^{(k+1)}\mathbf{BS}$, and $\mathbf{I}_{dr}$ denotes a $dr \times dr$ identity matrix.

It is obvious that there are two loops in the above optimization procedure. The inter loop occurs in the step of updating the matrix $\mathbf{V}$. This inter loop is iterated T_{in} times, where T_{in} is a hyperparameter determined by experiments. The outer loop updates the matrices $\mathbf{U}$ and $\mathbf{V}$, respectively and repeatedly. Since the optimization procedure includes a step of CNN-based denoising, convergence cannot be guaranteed mathematically. Therefore, the outer loop is iterated by a fixed number T_o. To accelerate the outer loop, we terminate the outer loop if $|\Phi(\mathbf{U}^{(k-1)},\mathbf{V}^{(k-1)})/\Phi(\mathbf{U}^{(k)},\mathbf{V}^{(k)}) - 1| < \varepsilon$. ε is a small parameter, which is set to 10^{-4}.

Besides, since the problem (10) is highly nonconvex, a good initialization is necessary for the optimization procedure. For simplicity, we directly initialize the matrices $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{W}$ based on $\mathbf{H}$ and summarize the proposed deep fusion method via PMF (DPMF) in Algorithm 1.

C. Deep Prior and the CNN-Based Gaussian Denoiser

This section introduces the CNN-based denoising solution for solving (17) and (18). It is obvious that two aspects have to be considered for the denoising solution, including the network design and the training strategy.

For network design, two requirements should be considered. First, the Gaussian denoiser should utilize the noise variance which is changed during the iteration and is known

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

in our framework. Second, the Gaussian denoiser should work perfectly at low noise variance since (λ_v/c) in (18) is usually very small. So far, there are many CNN-based Gaussian denoisers [34], [37], [38], [39]. However, only some of these denoising networks can satisfy the first requirement, e.g., FFDNet [39] and DnCNN [34], [38]. FFDNet introduces a noise level map as its input to utilize the known noise variance for denoising. Unfortunately, this denoising network tends to work better at high noise levels compared with DnCNN [38], [39]. DnCNN cannot directly utilize the noise variance [38]. Fortunately, Lin et al. [34] propose a data normalization scheme to utilize the noise variance based on DnCNN. In the proposed method, we directly adopt DnCNN used in [34] (see Fig. 1), since DnCNN can work well at low noise levels. However, the data normalization scheme proposed in [34] cannot be directly adopted in this work due to the nonnegativity constraint on $\mathbf{Q}_{i.}^{(n)}$ in (18). Thus, we propose a data normalization scheme as follows:

$$\mathbf{V}_{i.}^{(n+1)} = \sqrt{\frac{\lambda_v}{c}} \cdot \mathrm{DnCNN}\left(\left(\mathbf{Q}_{i.}^{(n)} - \min_j \mathbf{Q}_{i,j}^{(n)}\right)\sqrt{\frac{c}{\lambda_v}}\right). \quad (21)$$

For network training, two strategies are usually adopted, including the transfer learning strategy and the real-life HSI-based strategy [34]. The transfer learning strategy trains the CNN-based denoiser using gray images, whereas the real-life HSI-based strategy trains the CNN-based denoiser using data constructed from real-life HSIs. These two strategies try to generate 2-D image data, which contains various texture information. According to [34], it would be better to select the real-life HSI-based strategy since this strategy works better to remove the Gaussian noise of HSIs. Selections of network design and training strategy will be further discussed in Section IV.

IV. Experimental Results and Analysis

A. Testing Dataset

Four datasets are introduced to validate the proposed fusion method, including the CAVE dataset [13], the BGU ICVL dataset [14], the Indian Pines dataset, and the Washington DC Mall dataset. These four datasets only contain high-quality HSIs. Specifically, the CAVE and ICVL datasets are images captured on the ground. The CAVE dataset contains 32 HSI scenes with 31 spectral channel bands ranging from 400 to 700 nm at 10-nm intervals in wavelength [13]. The BGU ICVL dataset originally consists of 200 1392 × 1300 × 31 HSI scenes, covering the same wavelength as the CAVE dataset [14]. For simplicity, we randomly select 30 HSI scenes and resize each HSI for the test, as shown in Table I. The Indian Pines and Washington DC Mall datasets are remote sensing datasets. The Indian Pines dataset is an HSI cropped from an image captured by NASA's Airborne Visible Infrared Imaging Spectrometer (AVIRIS), which has 188 spectral bands after removing bands aggravated by water absorptions [22]. The Washington DC Mall dataset is an HSI cropped from an image captured by the HYDICE sensor, which has 191 spectral bands after removing noisy bands.

To evaluate the fusion method, we take the HSIs in the datasets as ground truths, low-resolution HSI (LR-HSI), and high-resolution MSI (HR-MSI) or RGB images are generated using a simulation tool provided by the MATLAB Hyperspectral Pansharpening Toolbox.[1] Specifically, the LR-HSI is generated by downsampling the origin high-resolution HSI at the scale of s after using a $(2s-1)\times(2s-1)$ Gaussian filter with the variance of $s^2/5.5452$, whereas the HR-MSI (or RGB image) is generated using a known spectral response matrix $\mathbf{R}$. For the CAVE and BGU ICVL datasets, we adopt the spectral sensing matrix of the Nikon D700 camera to generate RGB images. For the Indian Pines and Washington DC Mall datasets, we normalize the pixel values between 0 and 1. Then, we directly select six bands of the HSI at the wavelengths of 480, 560, 660, 830, 1650, and 2220 nm to simulate the HR-MSI acquired by the USGS/NASA Landsat 7 satellite. Based on the generated LR-HSIs and HR-MSIs (or RGB images), we further add white Gaussian noise to simulate the perturbations for three datasets, and let $\mathrm{SNR_h}$ and $\mathrm{SNR_m}$ denote the signal-to-noise ratios (SNRs) of HSI and MSI/RGB, respectively. The sizes of the original HSI, LR-HSI, and HR-MSI/RGB images are all shown in Table I for different datasets. Then, we take four quality assessment metrics used in [10] to evaluate the difference between the ground truth and the fusion result, including PSNR, SAM, ERGAS, and SSIM.

TABLE I
Images' Size of the Datasets Used in the Experiments

Dataset Name	Original HSI	LR-HSI	HR-MSI/RGB
CAVE [13]	512×512×31	$\frac{512}{s}\times\frac{512}{s}\times31$	512×512×3
ICVL [14]	512×512×31	$\frac{512}{s}\times\frac{512}{s}\times31$	512×512×3
Indian Pines	240×240×188	$\frac{240}{s}\times\frac{240}{s}\times188$	240×240×6
Washington DC	240×240×191	$\frac{240}{s}\times\frac{240}{s}\times191$	240×240×6

B. Implementation Details

1) Selection of Model Hyperparameters: For the selection of model hyperparameters, we choose DnCNN and FFDNet as the Gaussian CNN-Denoiser for the proposed DPMF method and train the CNN-Denoiser with 400 180 × 180 gray images following [38]. Then, we fixed $d = r = 10$ and run the proposed fusion method under different values of λ_u and λ_v. After fixing the values of λ_u and λ_v, we conduct the proposed fusion method under different values of r and d based on eight randomly selected HSIs of the CAVE dataset and show the mean PSNRs of eight fusion results in Figs. 2 and 3.

From Figs. 2(a) and 3(a), we can find that the PSNR of DPMF + DnCNN reaches a relatively good value when $0.8 \le \lambda_v \le 1$; the PSNR of DPMF + FFDNet reaches a relatively good value when $6 \le \lambda_v \le 8$. It is obvious that the selection of parameter λ_v is related to the CNN-based denoiser architecture. According to [34], the value of λ_v is also related to training strategy. Thus, we will further discuss the value of λ_v for the selection of CNN-based denoiser and training strategy. For simplicity, we let $\lambda_v = 1$ for DPMF + DnCNN and let $\lambda_v = 7$ for DPMF + FFDNet.

[1]The toolbox can be download at the website: https://openremotesensing.net/knowledgebase/hyperspectral-pansharpening-a-review/

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

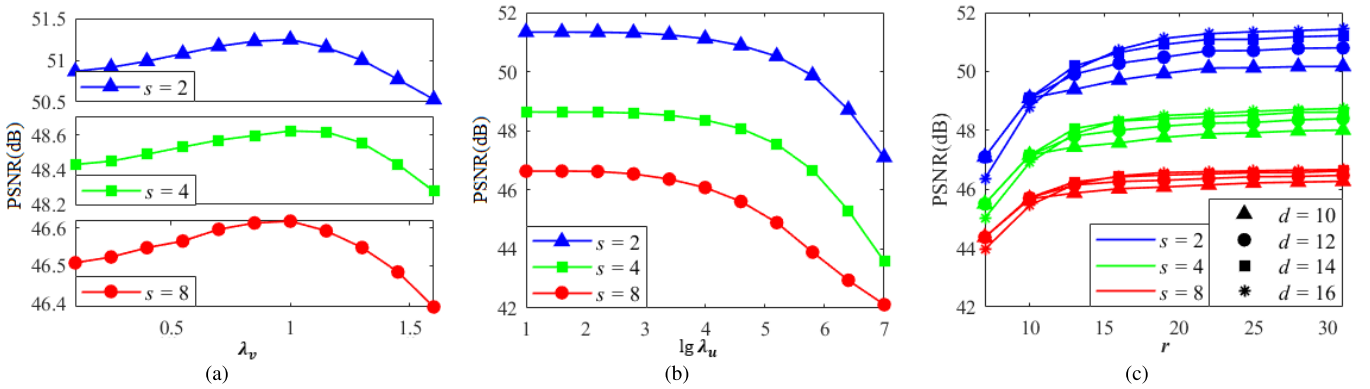


Fig. 2. Mean PSNR values of the DPMF fusion method for different settings of hyperparameters when using DnCNN denoiser trained with gray images. $\mathrm{SNR_m} = 40$ dB, $\mathrm{SNR_h} = 35$ dB, $T_{\mathrm{in}} = 2$, and $T_o = 150$. (a) PSNR values of λ_v when $d = 16, r = 25$, and $\lambda_u = 10^1$. (b) PSNR values of λ_u when $d = 16, r = 25$, and $\lambda_v = 1$. (c) PSNR values of d and r when $\lambda_u = 10^1$ and $\lambda_v = 1$.

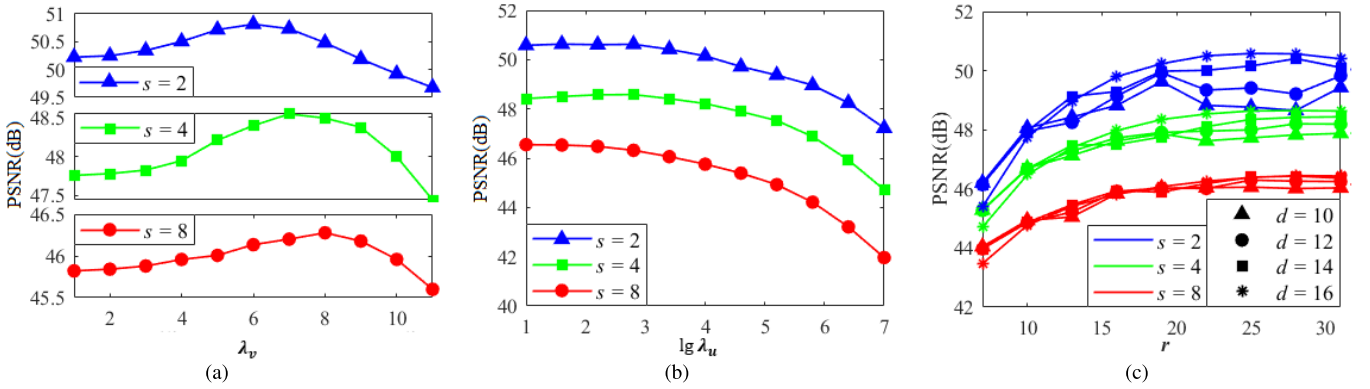


Fig. 3. Mean PSNR values of the DPMF fusion method for different settings of hyperparameters when using FFDNet denoiser trained with gray images. $\mathrm{SNR_m} = 40$ dB, $\mathrm{SNR_h} = 35$ dB, $T_{\mathrm{in}} = 2$, and $T_o = 150$. (a) PSNR values of λ_v when $d = 16, r = 25$, and $\lambda_u = 10^1$. (b) PSNR values of λ_u when $d = 16, r = 25$, and $\lambda_v = 7$. (c) PSNR values of d and r when $\lambda_u = 10^1$ and $\lambda_v = 7$.

From Figs. 2(b) and 3(b), we can find that the PSNR values of DPMF + DnCNN/FFDNet remain almost the same with the increase in λ_u when $\lambda_u \leq 10^3$, which implies that the CNN-Denoiser architecture is not related to the selection of λ_u. However, when $\lambda_u > 10^3$, the PSNR values of DPMF + DnCNN/FFDNet decrease with the increase in λ_u. Therefore, it is better to select a small value for λ_u. Considering (19) and (20), we let $\lambda_u = 10$ to guarantee that the matrix Σ_u is invertible.

From Figs. 2(c) and 3(c), it can be found that the DPMF + DnCNN/FFDNet performs better with the increase in r and d when $r \leq 25$ and $d \leq 14$. Unfortunately, the PSNR values of DPMF + DnCNN tend to be stable when $r > 25$ and $d > 14$; the PSNR values of DPMF + FFDNet decrease and show an unstable trend when $r > 25$. Thus, to ensure fusion performance, we set $r = 25$ and $d = 16$ for DPMF + DnCNN/FFDNet.

2) Selection of CNN-Denoiser Architecture: To select the best CNN-based denoiser architecture, we compare the fusion results of DPMF + DnCNN/FFDNet on the whole CAVE dataset. For fairness, we train the DnCNN and FFDNet denoisers with 400 180×180 gray images following [38] and adopt the carefully tuned hyperparameter settings. The evaluations of fusion results are shown in Table III. From Table III, we can find that the DPMF + FFDNet method shows worse fusion performance than that of the DPMF + DnCNN method.

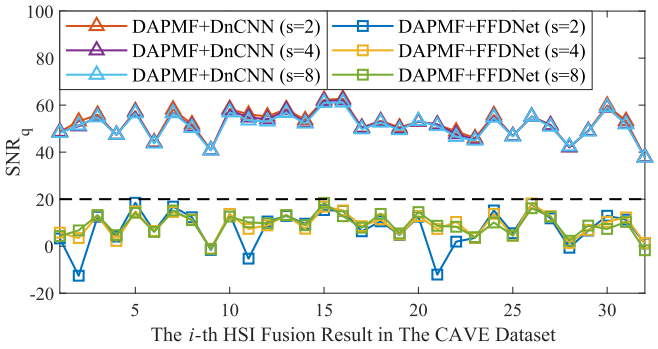


Fig. 4. Values of SNR_q for 32 HSI fusion results in the CAVE dataset.

To find the reason why the DPMF + DnCNN method shows better fusion performance than that of the DPMF + FFDNet method, we analyze the SNR of the input signal for the CNN-based denoiser architecture in the proposed DPMF fusion method. For DPMF + DnCNN, we approximately calculate the minimum SNR of the CNN input signal based on (21)

$$\mathrm{SNR}_q = \min_n 10 \lg \left[\frac{1}{MNr} \sum_{i,k} \left(\mathbf{V}_{i,k}^{(n+1)} \right)^2 \cdot \frac{c}{\lambda_v} \right]. \tag{22}$$

Similarly, for DPMF + FFDNet, we approximately calculate the minimum SNR of the CNN input signal based on (22)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

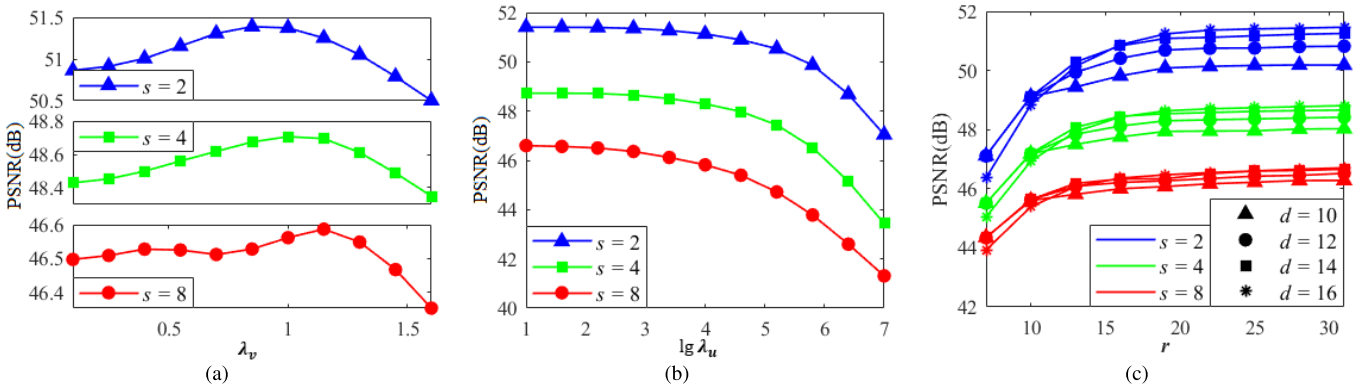


Fig. 5. Mean PSNR values of the DPMF fusion method for different settings of hyperparameters when using DnCNN denoiser trained with images constructed from real-life HSIs. $\text{SNR}_\text{m} = 40$ dB, $\text{SNR}_\text{h} = 35$ dB, $T_\text{in} = 2$, and $T_o = 150$. (a) PSNR values of λ_v when $d = 16$, $r = 25$, and $\lambda_u = 10^1$. (b) PSNR values of λ_u when $d = 16$, $r = 25$, and $\lambda_v = 1$. (c) PSNR values of d and r when $\lambda_u = 10^1$ and $\lambda_v = 1$.

TABLE II
OUR PREVIOUS MF-BASED FUSION WORKS WITH DIFFERENT REGULARIZATIONS

Method	Gaussian	Non-negativity	Deep prior
BSDMF [5]	✓		
NMF-DPR [12]		✓	✓
DPMF	✓		✓

by replacing $\mathbf{V}_{i.}^{(n+1)}$ with the output signal of FFDNet. It is obvious that SNR_q is related to the manually tuned hyperparameter λ_v. To fairly compare DPMF + FFDNet and DPMF + DnCNN, we adopt the carefully tuned hyperparameters settings introduced in Section IV-B1 and calculate all values of SNR_q for 32 HSI fusion result on the CAVE dataset using DPMF + FFDNet and DPMF + DnCNN, respectively. The results are shown in Fig. 4.

From Fig. 4, we can find that $\text{SNR}_q \leq 20$ dB for DPMF + FFDNet and $\text{SNR}_q \gg 20$ dB for DPMF + DnCNN when using the carefully tuned hyperparameter λ_v to ensure good fusion performance. These results are related to the best work condition of the CNN-based denoiser. Specifically, according to [39], FFDNet outperforms DnCNN when the noise level is higher than 25, i.e., $\sigma \geq 0.1$ and $\text{SNR} \leq 20$ dB. To ensure the fusion performance of the proposed DAPMF method, the hyperparameter λ_v should be carefully tuned to guarantee that FFDNet and DnCNN work under the best SNR range (e.g., $\text{SNR}_q \leq 20$ dB for DPMF + FFDNet) in every iteration of the outer loop. However, with the increase in iteration times, the SNR of the input signal for the CNN-based denoiser increases. In this case, FFDNet works out of the best SNR range, whereas DnCNN still works excellently at high SNR (i.e., with a low noise variance). Therefore, DnCNN is more suitable than FFDNet to solve (17) for the proposed DPMF method.

3) Selection of Training Strategy: To select the best training strategy, we further conduct the DPMF + DnCNN method using the real-life HSI-based training strategy [34] under different values of λ_u, λ_v, r, and d. These results are shown in Fig. 5. It can be found from Fig. 5 that the PSNR reaches a relatively good value when $\lambda_v = 1$, $\lambda_u = 10$, $d = 16$, and $r = 25$. Besides, Fig. 5 shares the similar changing rules as that of Fig. 2 except Fig. 5(c), which implies that the selection of training strategy only influences the tuning of λ_v.

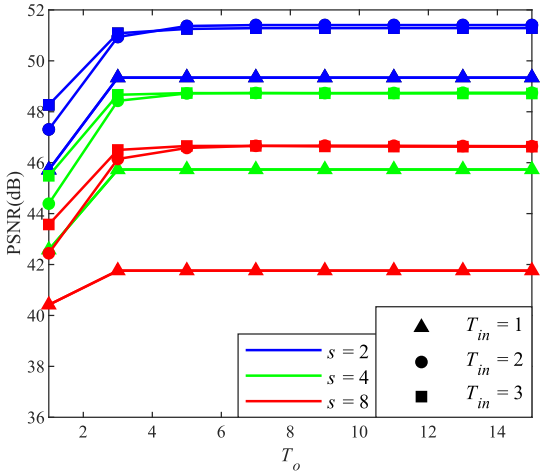


Fig. 6. Mean PSNR values of DPMF + DnCNN under different values of T_in and T_o. $\text{SNR}_\text{m} = 40$ dB, $\text{SNR}_\text{h} = 35$ dB, $d = 16$, $r = 25$, $\lambda_u = 10$, and $\lambda_v = 1$.

Based on the above carefully tuned hyperparameter settings, we compare the fusion results of DPMF + DnCNN based on different test datasets using different training strategies. The results are shown in Tables III–VI. It can be found that DPMF + DnCNN* performs slightly better on the CAVE and ICVL datasets, whereas the performance of DPMF + DnCNN is very close to that of DPMF + DnCNN* on the Remote Sensing datasets. Nevertheless, the architecture of DPMF + DnCNN is not sensitive to training strategy since its fusion performance is very close under different training strategies.

4) Selection of Iteration Numbers: To tune the iteration numbers T_in and T_o, we conduct the DPMF + DnCNN method using the real-life HSI-based training strategy proposed in [34] based on eight HSIs randomly selected from the CAVE dataset under different values of T_in and T_o. The results are shown in Fig. 6. From Fig. 6, we can find that the PSNR reaches a better convergence value when $T_\text{in} \geq 2$. Besides, thanks to (14) based on the iPALM algorithm, the proposed fusion method can reach convergence quickly when $T_o \leq 7$. Thus,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

TABLE III
FUSION RESULTS ON THE CAVE DATASET ($SNR_m = 40$ dB AND $SNR_h = 35$ dB)

Scale	Methods	Evaluation Metrics (Mean±Std.Dev)				Avr. Time
		PSNR	SAM	ERGAS	SSIM	
$s = 2$	BSDMF [5]	48.08±3.50	3.64±1.19	3.113±1.660	0.9928±0.0047	15
	NMF-DPR [12]	48.74±3.88	3.38±1.01	2.792±1.641	0.9935±0.0039	68
	LRTA [10]	48.07±3.57	5.35±2.15	2.602±1.064	0.9878±0.0115	71
	CSTF [8]	48.53±3.79	4.42±1.64	2.776±1.626	0.9915±0.0063	80
	CNN-FUS [22]	47.05±3.73	3.23±1.02	3.224±1.789	0.9900±0.0087	39
	Rec-HSISR [40]	45.18±2.34	3.66±1.34	3.940±1.996	0.9917±0.0036	982
	DPMF+FFDNet[1]	48.41±4.35	3.69±1.27	2.727±1.371	0.9915±0.0097	9
	DPMF+DnCNN[1]	**49.82±3.72**	**3.00±0.97**	**2.451±1.261**	**0.9948±0.0030**	**7**
	DPMF+DnCNN*[2]	**49.98±3.73**	**2.95±0.96**	**2.441±1.262**	**0.9949±0.0030**	**7**
$s = 4$	BSDMF [5]	45.36±3.65	4.38±1.34	2.157±1.088	0.9885±0.0073	14
	NMF-DPR [12]	46.83±3.94	4.20±1.23	1.795±0.958	0.9904±0.0054	33
	LRTA [10]	46.31±3.56	5.88±2.15	1.697±0.725	0.9845±0.0110	90
	CSTF [8]	46.57±3.69	4.97±1.66	1.812±0.940	0.9883±0.0072	80
	CNN-FUS [22]	43.51±3.67	4.16±1.45	2.419±1.232	0.9799±0.0172	38
	Rec-HSISR [40]	43.85±2.97	3.95±1.38	2.283±1.141	0.9891±0.0055	974
	DPMF+FFDNet[1]	46.84±3.76	4.54±1.63	1.760±0.852	0.9885±0.0091	**10**
	DPMF+DnCNN[1]	**47.19±3.85**	**3.88±1.23**	**1.732±0.882**	**0.9912±0.0054**	**9**
	DPMF+DnCNN*[2]	**47.29±3.88**	**3.73±1.17**	**1.724±0.884**	**0.9915±0.0053**	11
$s = 8$	BSDMF [5]	43.03±3.83	5.44±1.78	1.404±0.733	0.9826±0.0129	14
	NMF-DPR [12]	45.17±4.01	5.07±1.56	1.089±0.564	**0.9877±0.0068**	28
	LRTA [10]	**45.60±3.70**	5.11±1.50	**0.989±0.466**	0.9863±0.0085	214
	CSTF [8]	44.83±3.59	5.72±1.96	1.112±0.514	0.9840±0.0093	80
	CNN-FUS [22]	41.56±3.46	5.86±2.19	1.492±0.709	0.9699±0.0208	37
	Rec-HSISR [40]	43.12±2.83	**4.14±1.35**	1.245±0.609	**0.9874±0.0063**	1011
	DPMF+FFDNet[1]	44.85±3.84	5.24±1.61	1.101±0.509	0.9846±0.0105	**10**
	DPMF+DnCNN[1]	**45.30±4.01**	4.82±1.52	**1.075±0.552**	0.9869±0.0091	**13**
	DPMF+DnCNN*[2]	45.30±4.06	**4.70±1.50**	1.077±0.551	0.9870±0.0092	15

[1] FFDNet and DnCNN are both trained with 400 180×180 gray images following [38].
[2] DnCNN* is trained with 2D images constructed from 20 real-life HSIs of the Harvard dataset following [34].

to guarantee the convergence and the fusion performance, we let $T_{\text{in}} = 2$ and $T_o = 15$.

C. Comparison With Other Fusion Methods

In this section, six fusion methods are introduced for comparison, including BSDMF [5], NMF-DPR[2] [12], LRTA[3] [10], CSTF[4] [8], CNN-FUS[5] [22], and Rec-HSISR[6] [40]. Specifically, BSDMF and NMF-DPR are our previous matrix factorization (MF)-based fusion works with different prior regularizations, which is shown in Table II. We compare the proposed DPMF method with BSDMF and NMF-DPR to validate the effectiveness of DPR under MF-based modeling. LRTA and CSTF are tensor-decomposition-based fusion methods, whereas CNN-FUS is a subspace-based fusion method with DPR. These three non-MF-based methods are compared with our MF-based methods to validate the advantage of MF-based methods. Rec-HSISR is an unsupervised deep-learning-based fusion method. We introduce Rec-HSISR for comparison since the unsupervised deep-learning-based method is very popular in recent years. In the following passage, all comparison methods are run on the same computer with Intel Core 3.6-GHz i7-11700 CPU and 16-GB RAM. All computations of neural networks are accelerated by Nvidia RTX 3080 GPU. Based on the same experiment condition, we will discuss the parameter settings and the fusion performance of all comparison methods based on different test datasets.

1) Comparison on the CAVE and ICVL Datasets: For fairness, we carefully tune all hyperparameters of all comparison fusion methods based on the CAVE dataset. Specifically, for BSDMF, since this fusion method shares the same PMF model as that of the proposed DPMF method [5], we let $d = 16$ and $r = 25$ for the BSDMF method following the setting of d and r in the proposed fusion method. For NMF-DPR, since this fusion method is also an MF-based method, we let $r = 16$ for NMF-DPR so that this method shares the same matrix rank as that of the proposed method. For LRTA, we directly adopt the parameter setting proposed in [10]. To guarantee convergence, we set the maximum iteration number to 120. For CSTF, the hyperparameters, n_w, n_h, and n_s, are related to the size of HSI in the fusion method. Since the size of HSI in the CAVE

[2]The source code of BSDMF and NMF-DPR can be downloaded at the website: https://github.com/KevinBHLin
[3]The source code of LRTA can be download at the website: https://openremotesensing.net/
[4]The source code of CSTF can be download at the website: https://github.com/renweidian/CSTF
[5]The source code of CNN-FUS can be download at the website: https://github.com/renweidian/CNN-FUS
[6]The source code of Rec-HSISR can be download at the website: https://github.com/JiangtaoNie/Rec_HSISR_PixAwaRefin

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
FUSION RESULTS ON THE ICVL DATASET ($SNR_m = 40$ dB AND $SNR_h = 35$ dB)

Scale	Methods	Evaluation Metrics (Mean±Std.Dev)				Avr. Time
		PSNR	SAM	ERGAS	SSIM	
$s = 2$	BSDMF [5]	53.43±2.87	0.78±0.12	1.016±0.234	0.9964±0.0015	12
	NMF-DPR [12]	53.25±2.98	0.92±0.10	0.833±0.125	0.9963±0.0016	24
	LRTA [10]	48.04±3.06	1.71±0.15	1.407±0.067	0.9858±0.0068	70
	CSTF [8]	50.37±2.89	1.49±0.11	1.160±0.052	0.9896±0.0049	72
	CNN-FUS [22]	49.12±2.90	0.83±0.16	1.280±0.256	0.9914±0.0037	31
	Rec-HSISR [40]	49.65±2.37	1.00±0.33	1.294±0.448	0.9957±0.0017	966
	DPMF+FFDNet[1]	47.78±6.44	1.14±0.44	1.868±0.870	0.9796±0.0216	**4**
	DPMF+DnCNN[1]	**55.54±3.31**	**0.56±0.11**	**0.689±0.134**	**0.9981±0.0009**	6
	DPMF+DnCNN*[2]	**55.42±3.20**	**0.56±0.11**	**0.696±0.122**	**0.9981±0.0009**	**5**
$s = 4$	BSDMF [5]	51.57±2.97	0.96±0.18	0.711±0.194	0.9953±0.0019	11
	NMF-DPR [12]	51.64±2.85	1.11±0.14	0.528±0.079	0.9958±0.0018	22
	LRTA [10]	47.16±3.07	1.90±0.18	0.804±0.076	0.9871±0.0061	95
	CSTF [8]	48.11±2.94	1.96±0.21	0.788±0.058	0.9855±0.0065	71
	CNN-FUS [22]	44.44±2.25	1.02±0.25	1.107±0.263	0.9756±0.0086	30
	Rec-HSISR [40]	48.83±2.60	1.09±0.27	0.706±0.207	0.9947±0.0022	975
	DPMF+FFDNet[1]	52.75±3.84	0.83±0.20	0.511±0.130	0.9968±0.0016	12
	DPMF+DnCNN[1]	**53.42±3.41**	**0.81±0.15**	**0.446±0.095**	**0.9973±0.0013**	**6**
	DPMF+DnCNN*[2]	**53.77±3.37**	**0.74±0.13**	**0.432±0.091**	**0.9974±0.0013**	**7**
$s = 8$	BSDMF [5]	50.15±3.83	1.15±0.26	0.449±0.134	0.9946±0.0023	**11**
	NMF-DPR [12]	50.57±2.95	1.22±0.19	0.320±0.050	0.9960±0.0017	20
	LRTA [10]	48.90±3.72	1.45±0.32	0.381±0.073	0.9941±0.0029	210
	CSTF [8]	47.19±3.44	2.07±0.22	0.452±0.061	0.9866±0.0078	68
	CNN-FUS [22]	42.57±1.87	1.59±0.44	0.700±0.180	0.9673±0.0101	30
	Rec-HSISR [40]	48.09±3.51	1.16±0.35	0.393±0.149	0.9936±0.0047	1012
	DPMF+FFDNet[1]	51.88±3.44	**0.93±0.18**	0.289±0.065	0.9965±0.0017	12
	DPMF+DnCNN[1]	**52.46±3.35**	0.95±0.16	**0.254±0.052**	**0.9971±0.0014**	**11**
	DPMF+DnCNN*[2]	**52.56±3.34**	**0.90±0.15**	**0.251±0.054**	**0.9971±0.0014**	**8**

[1] FFDNet and DnCNN are both trained with 400 180×180 gray images following [38].
[2] DnCNN* is trained with 2D images constructed from 20 real-life HSIs of the Harvard dataset following [34].

dataset is larger than the test HSI in [8], we tune the parameters and let $n_w = n_h = 480$ and $n_s = 16$. For CNN-FUS, the hyperparameter L denotes the column size of the subspace matrix obtained by SVD. Similar to Figs. 2(c) and 5(c), we find that the PSNR increases with the increase in L on the CAVE dataset. Since L is equivalent to the hyperparameter d of the proposed fusion method, we let $L = 16$ to guarantee that the CNN-FUS method is compared with the proposed fusion method under the same spectral subspace. For Rec-HSISR, we find that this method performs badly on the test datasets when directly adopting the parameter setting given by Wei et al. [40]. By tuning the hyperparameters, we find that this method shows the best performance when the learning rate is set to 10^{-2}. Based on the above parameter settings, the results of all comparison methods are shown in Tables III and IV. Visualization examples of fusion results are displayed in Figs. 7 and 8.

From the results in Tables III and IV, we can find that the proposed DPMF + DnCNN and DPMF + DnCNN* shows the best fusion performance under all evaluation metrics in most cases. Besides, DPMF + DnCNN and DPMF + DnCNN* cost less computation time since DPMF + DnCNN and DPMF + DnCNN* reach convergence within a few iterations according to Fig. 6, and the computation of CNN adopted in the proposed DPMF is accelerated by GPU.

As for the comparison methods, they perform worse for different reasons. For BSDMF, although this method shares the same PMF model as that of the DPMF method, the abundance matrix is regularized by Gaussian priors, which cannot utilize the texture information to enhance the fusion performance like DPR. Thus, it performs worse than NMF-DPR and the proposed DPMF. However, BSDMF is very fast since it costs less computation time without GPU acceleration compared with most methods in Tables III and IV. For NMF-DPR, although this method also adopts DPR, it shows worse fusion performance compared with the proposed DPMF method. The results of NMF-DPR prove that nonnegativity constraint on factor matrices does not bring in better fusion performance for MF-based fusion methods compared with Gaussian prior. Moreover, although NMF-DPR is accelerated by GPU, NMF-DPR requires more computation time than the proposed DPMF. The main reason is that the convergence speed of NMF-DPR is slow due to the nonnegativity constraint. For CSTF and LRTA, the low rank tensor factorization model is designed to model the texture and channel information priors laid in HSI. However, we can find from Figs. 7 and 8 that this model can easily introduce stripe noise along horizontal and vertical directions, which imply that spatial texture cannot be fully utilized using the low rank tensor factorization model. Moreover, both LRTA and CSTF

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

TABLE V
FUSION RESULTS ON THE INDIAN PINES DATASET

s	Methods	SNR_m=40dB, SNR_h=35dB				SNR_m=35dB, SNR_h=30dB				SNR_m=30dB, SNR_h=25dB				Avr.
		PSNR	SAM	ERGAS	SSIM	PSNR	SAM	ERGAS	SSIM	PSNR	SAM	ERGAS	SSIM	Time
2	BSDMF [5]	53.82	0.80	2.963	0.9971	51.20	1.13	3.289	0.9940	48.84	1.61	3.647	0.9871	9
	NMF-DPR [12]	55.26	0.78	2.220	0.9973	51.72	1.12	2.814	0.9942	49.11	1.69	3.289	0.9866	33
	LRTA [10]	49.89	1.64	2.911	0.9897	46.77	2.62	3.470	0.9753	42.90	4.37	4.840	0.9372	40
	CSTF [8]	**56.84**	0.88	**1.547**	0.9971	53.59	1.39	1.965	0.9930	49.79	2.34	2.681	0.9812	18
	CNN-FUS [22]	54.95	0.85	1.610	0.9953	53.84	1.21	1.786	0.9931	51.32	2.10	2.259	0.9843	107
	Rec-HSISR [40]	47.74	1.33	4.894	0.9882	47.82	1.32	4.878	0.9884	47.07	1.49	4.988	0.9864	247
	DPMF+FFDNet[1]	43.98	2.06	5.088	0.9648	43.89	2.18	5.143	0.9639	43.73	2.46	5.213	0.9616	**1**
	DPMF+DnCNN[1]	56.09	**0.77**	1.613	**0.9976**	**54.42**	**0.94**	**1.681**	**0.9962**	**51.71**	**1.24**	**1.974**	**0.9920**	**3**
	DPMF+DnCNN*[2]	**56.53**	**0.75**	**1.576**	**0.9977**	**54.21**	**0.97**	**1.671**	**0.9959**	**51.96**	**1.21**	**1.917**	**0.9927**	**3**
4	BSDMF [5]	50.86	1.08	2.090	0.9953	49.17	1.36	2.171	0.9927	47.19	1.87	2.319	0.9862	9
	NMF-DPR [12]	52.20	0.92	1.728	0.9967	49.37	1.34	1.962	0.9929	46.95	2.00	2.143	0.9836	17
	LRTA [10]	45.89	2.13	2.464	0.9783	44.41	3.27	2.600	0.9672	42.17	4.32	2.946	0.9484	58
	CSTF [8]	54.52	1.17	0.947	0.9952	52.44	1.68	1.084	0.9911	48.73	2.72	1.524	0.9786	17
	CNN-FUS [22]	52.35	1.09	1.083	0.9909	51.54	1.36	1.169	0.9889	49.01	2.23	1.575	0.9826	106
	Rec-HSISR [40]	47.31	1.40	2.485	0.9869	45.50	1.91	2.663	0.9816	46.38	**1.62**	2.556	0.9842	286
	DPMF+FFDNet[1]	52.75	1.14	1.079	0.9944	52.43	**1.15**	1.075	0.9948	48.13	1.85	1.548	0.9837	4
	DPMF+DnCNN[1]	**54.77**	**0.90**	**0.936**	**0.9970**	**53.19**	**1.15**	**1.012**	**0.9954**	**50.02**	**1.65**	**1.323**	**0.9908**	**3**
	DPMF+DnCNN*[2]	**54.76**	**0.89**	**0.945**	**0.9970**	**53.12**	**1.10**	**1.032**	**0.9955**	**49.77**	**1.65**	**1.352**	**0.9904**	**3**
8	BSDMF [5]	49.64	1.28	1.107	0.9947	48.49	1.53	1.136	0.9929	46.57	2.05	1.206	0.9873	9
	NMF-DPR [12]	50.56	1.17	0.947	0.9954	48.63	1.48	1.035	0.9923	46.96	1.96	1.093	0.9865	19
	LRTA [10]	41.92	3.52	1.738	0.9533	41.70	3.58	1.742	0.9527	41.30	3.81	1.762	0.9502	138
	CSTF [8]	52.42	1.17	0.572	0.9947	50.33	1.99	0.654	0.9881	47.23	3.20	0.965	0.9767	17
	CNN-FUS [22]	50.23	1.40	0.668	0.9862	49.42	1.59	0.758	0.9818	47.98	2.14	0.875	0.9858	106
	Rec-HSISR [40]	46.37	1.55	1.282	0.9831	46.01	1.64	1.303	0.9816	45.45	1.90	1.324	0.9800	382
	DPMF+FFDNet[1]	52.15	1.35	0.541	0.9932	49.60	1.65	0.690	0.9881	46.47	2.29	0.949	0.9748	5
	DPMF+DnCNN[1]	**53.57**	**0.99**	**0.535**	**0.9967**	**50.84**	**1.38**	**0.639**	**0.9940**	**48.65**	**1.86**	**0.786**	**0.9907**	**4**
	DPMF+DnCNN*[2]	**53.55**	**0.99**	**0.529**	**0.9967**	**50.79**	**1.37**	**0.637**	**0.9940**	**48.60**	**1.79**	**0.784**	**0.9907**	**3**

[1] FFDNet and DnCNN are both trained with 400 180×180 gray images following [38].
[2] DnCNN* is trained with 2D images constructed from 20 real-life HSIs of the Harvard dataset following [34].

require more computation time compared with MF-based methods due to the high computational cost of tensor factorization without GPU acceleration. Besides, LRTA is reported to be better than CSTF [10], but we can find from Tables III and IV that LRTA only works better on the CAVE and ICVL datasets when $s = 8$, and LRTA requires more computational cost. As for CNN-FUS, different from the proposed DPMF method, CNN-FUS removes the matrix $\mathbf{U}$ in (6), i.e., the spectral matrix is obtained by SVD instead of being estimated from the fusion model [22]. Although deep prior is also introduced to regularize the coefficients, we can find from Tables III and IV that CNN-FUS performs worse than the proposed DPMF method, which demonstrates the effectiveness of PMF model adopted by the proposed DPMF method. Besides, we find that the convergence speed of CNN-FUS is very slow. Thus, CNN-FUS requires more computation time compared with most methods in Tables III and IV despite that it is accelerated by GPU. For Rec-HSISR, this method costs a lot of computation time with GPU acceleration but brings in plain fusion performance. In our experiments, we find that Rec-HSISR is sensitive to the learning rate, and this method is sometimes overfitting. Although the unsupervised deep learning fusion method is very promising due to its strong nonlinear fitting ability for modeling the spectral and spatial relation laid in a fusion image, this kind of method is still under development in recent years.

2) Comparison on the Remote Sensing Datasets: We test the proposed DPMF method on the Indian Pines and Washington DC Mall remote sensing datasets using the parameter setting introduced in Section IV-B. For fair comparisons, we still adopt the hyperparameter settings of all comparison methods used on the CAVE dataset except CSTF since the hyperparameters of CSTF are related to the spatial size of HSI. According to [8], we let $n_w = n_h = 240$ and $n_s = 16$ for CSTF. Based on the above parameter settings, the results are shown in Tables V and VI. Visualization examples of fusion results are displayed in Figs. 9 and 10.

From Table V, we can find that the proposed DPMF method still outperforms other comparison methods in most cases under different evaluation metrics and different SNRs, which demonstrates that the proposed DPMF method has high generalization ability without extra training processes. Thanks to the DPR and the iPALM algorithm, the CNN adopted in DPMF only requires a single training process based on a 2-D image dataset, and then, DPMF can be utilized to excellently fuse different types of HSI and MSI.

The other comparison methods have similar trends of computational cost to the results of the CAVE and ICVL datasets. However, some of them show different fusion performances from the results of the CAVE and ICVL datasets. For BSDMF and NMF-DPR, these two methods have similar performance to the results of the CAVE and ICVL datasets, i.e., they still

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
FUSION RESULTS ON THE WASHINGTON DC MALL DATASET

s	Methods	SNR_m=40dB, SNR_h=35dB				SNR_m=35dB, SNR_h=30dB				SNR_m=30dB, SNR_h=25dB				Avr.
		PSNR	SAM	ERGAS	SSIM	PSNR	SAM	ERGAS	SSIM	PSNR	SAM	ERGAS	SSIM	Time
2	BSDMF [5]	50.68	0.65	0.689	0.9954	47.54	0.91	0.983	0.9905	43.84	1.36	1.510	0.9776	9
	NMF-DPR [12]	51.36	0.66	0.646	0.9960	47.71	0.98	0.967	0.9910	44.71	1.44	1.390	0.9821	10
	LRTA [10]	42.08	2.00	2.132	0.9588	38.63	2.90	2.857	0.9231	34.71	4.60	4.349	0.8355	42
	CSTF [8]	51.75	0.74	0.634	0.9956	47.46	1.22	1.040	0.9881	42.66	2.21	1.842	0.9648	19
	CNN-FUS [22]	52.62	**0.61**	0.563	0.9967	49.17	1.04	0.870	0.9919	44.30	1.96	1.574	0.9744	113
	Rec-HSISR [40]	48.51	1.14	1.209	0.9882	47.73	1.19	1.273	0.9866	45.52	1.37	1.486	0.9831	246
	DPMF+FFDNet[1]	48.32	0.97	0.948	0.9919	42.26	2.55	3.587	0.9001	41.56	2.73	3.633	0.8973	4
	DPMF+DnCNN[1]	**54.14**	**0.62**	**0.554**	**0.9971**	**51.46**	**0.81**	**0.737**	**0.9949**	**48.40**	**1.11**	**1.031**	**0.9902**	**2**
	DPMF+DnCNN*[2]	**54.11**	0.63	**0.561**	**0.9971**	**51.35**	**0.83**	**0.750**	**0.9947**	**48.23**	**1.14**	**1.075**	**0.9895**	**2**
4	BSDMF [5]	49.26	0.75	0.410	0.9943	46.53	1.03	0.554	0.9894	42.73	1.54	0.858	0.9749	9
	NMF-DPR [12]	50.11	0.75	0.372	0.9952	46.29	1.16	0.572	0.9885	42.42	1.82	0.891	0.9727	14
	LRTA [10]	40.96	3.02	1.797	0.8971	35.83	3.66	2.098	0.8696	34.72	5.26	2.473	0.8263	61
	CSTF [8]	50.45	0.90	0.382	0.9943	46.57	1.39	0.581	0.9866	41.81	2.41	1.004	0.9614	19
	CNN-FUS [22]	51.35	**0.65**	0.324	0.9960	46.06	1.62	0.790	0.9781	43.01	2.15	0.886	0.9707	112
	Rec-HSISR [40]	46.62	1.22	0.677	0.9855	46.71	1.30	0.687	0.9846	45.29	**1.39**	0.753	0.9814	284
	DPMF+FFDNet[1]	49.03	0.93	0.449	0.9934	46.29	1.13	0.582	0.9894	38.08	4.25	2.851	0.7789	6
	DPMF+DnCNN[1]	**52.93**	**0.73**	**0.315**	**0.9964**	**50.40**	**0.96**	**0.414**	**0.9941**	**47.34**	**1.39**	**0.593**	**0.9882**	**3**
	DPMF+DnCNN*[2]	**53.07**	**0.73**	**0.314**	**0.9965**	**50.49**	**0.97**	**0.417**	**0.9940**	**47.24**	**1.35**	**0.593**	**0.9880**	**2**
8	BSDMF [5]	48.42	0.84	0.226	0.9943	45.55	1.17	0.310	0.9890	41.59	1.77	0.490	0.9740	9
	NMF-DPR [12]	48.85	0.85	0.213	0.9940	45.47	1.22	0.311	0.9876	43.20	**1.66**	0.405	0.9798	20
	LRTA [10]	37.61	4.04	1.267	0.8348	37.24	4.09	1.274	0.8338	34.94	5.87	1.462	0.8158	146
	CSTF [8]	47.92	1.28	0.266	0.9901	44.95	1.80	0.363	0.9819	40.82	2.74	0.563	0.9604	18
	CNN-FUS [22]	41.15	2.91	0.808	0.9268	46.09	1.37	0.313	0.9870	42.15	2.38	0.495	0.9708	112
	Rec-HSISR [40]	45.95	1.35	0.355	0.9833	44.50	1.51	0.402	0.9795	42.78	1.91	0.500	0.9688	471
	DPMF+FFDNet[1]	48.14	1.11	0.271	0.9903	45.32	1.29	0.325	0.9878	36.17	5.38	1.751	0.7290	6
	DPMF+DnCNN[1]	**51.82**	**0.81**	**0.175**	**0.9958**	**49.31**	**1.11**	**0.234**	**0.9928**	**45.93**	1.71	**0.353**	**0.9854**	**3**
	DPMF+DnCNN*[2]	**51.98**	**0.80**	**0.172**	**0.9960**	**49.16**	**1.12**	**0.238**	**0.9926**	**45.98**	**1.69**	**0.354**	**0.9853**	**3**

[1] FFDNet and DnCNN are both trained with 400 180×180 gray images following [38].
[2] DnCNN* is trained with 2D images constructed from 20 real-life HSIs of the Harvard dataset following [34].

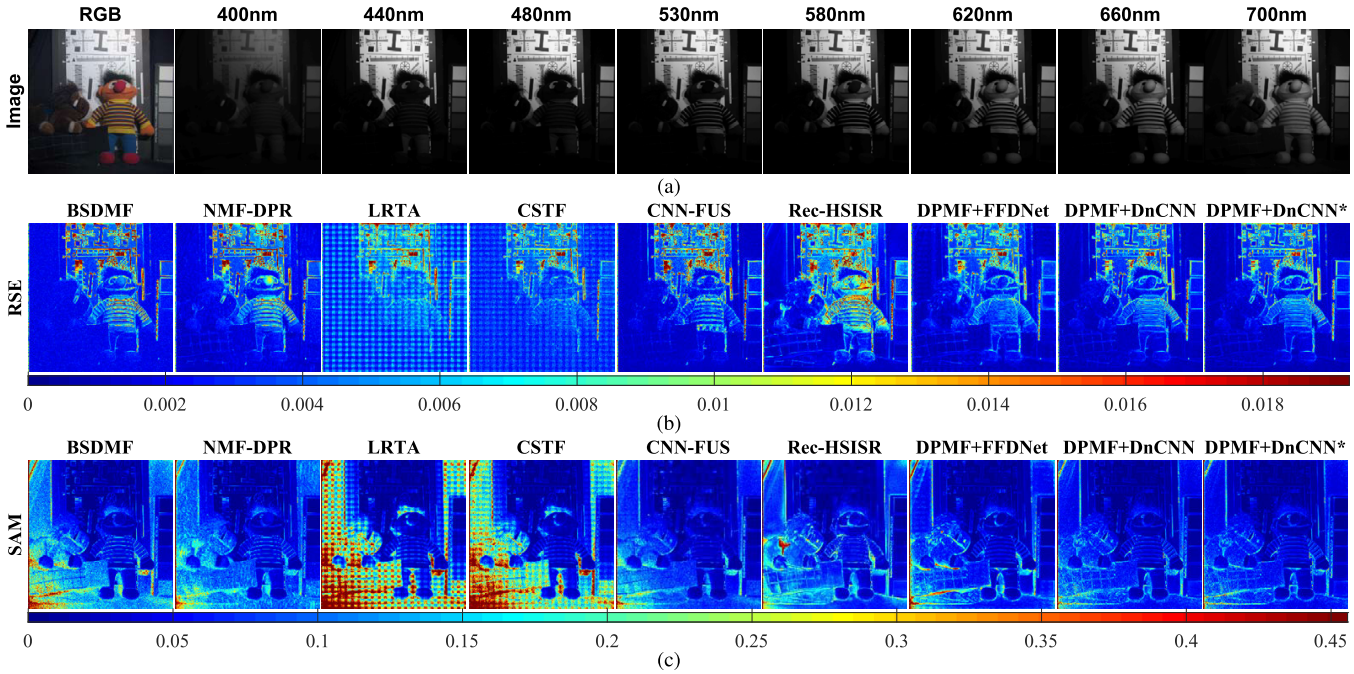


Fig. 7. Fusion result examples of the CAVE dataset when $s = 2$. (a) RGB and HSI. (b) Root square error. (c) Spectral angle maps.

perform worse than DPMF and require more computation time compared with DPMF. However, these two methods are very stable for the following reason: the hyperparameters of BSDMF are all automatically learned from the Bayesian inference process; NMF-DPR has only one hyperparameter (i.e., the column size of spectral matrix r), which is not very

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

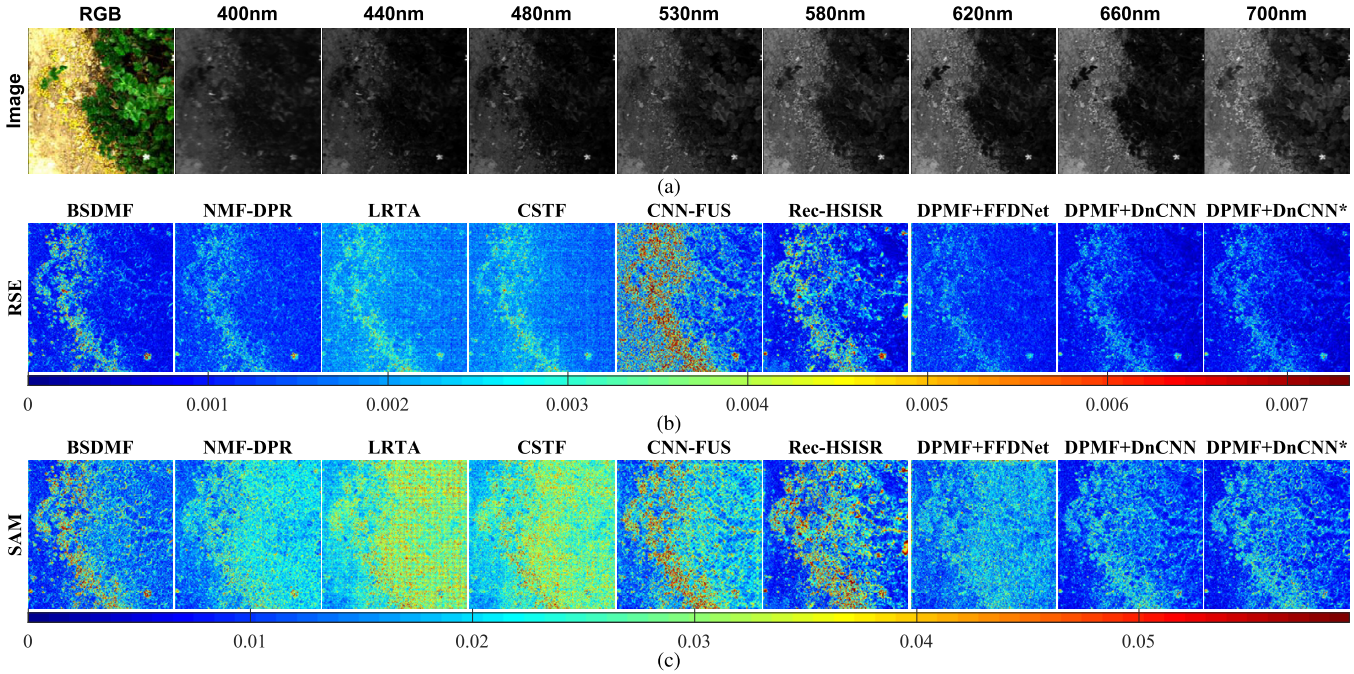


Fig. 8. Fusion result examples of the ICVL dataset when $s = 4$. (a) RGB and HSI. (b) Root square error. (c) Spectral angle maps.

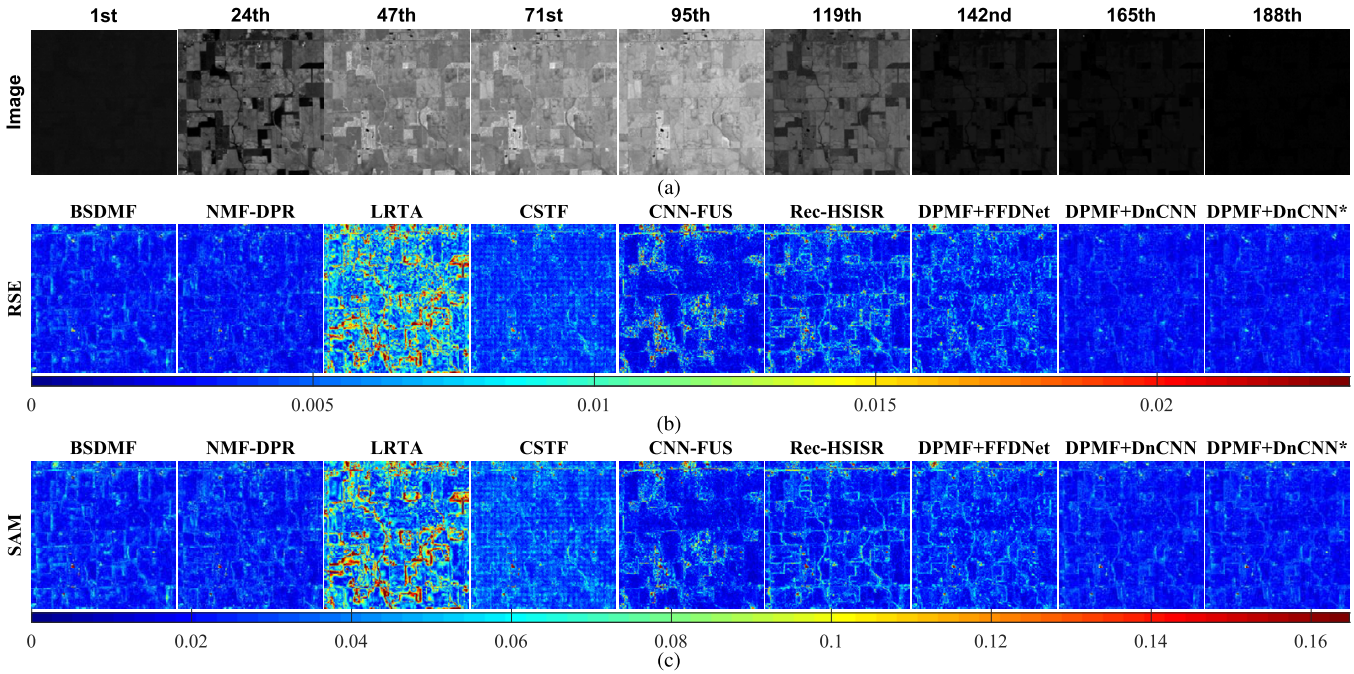


Fig. 9. Fusion results of Indian Pines when $s = 8$ ($\text{SNR}_m = 35$ dB and $\text{SNR}_h = 30$ dB). (a) HSI. (b) Root square error. (c) Spectral angle maps.

sensitive to different datasets. For LRTA, it shows the worst performance on two remote sensing datasets in most cases. The results of LRTA imply that the setting of hyperparameters given in [10] is not suitable for the simulation scenarios given in Section IV-A. For CSTF, this method shows good fusion performance on two remote sensing datasets, and its fusion performance is only secondary to that of the proposed DPMF fusion method on the Indian Pines dataset. Although LRTA is reported better than CSTF, we find from Tables V and VI that CSTF outperforms LRTA and requires a less computational cost on two remote sensing datasets. For CNN-FUS, its fusion performance decreases sharply with the increase in s on the CAVE and ICVL datasets. However, it shows relatively good fusion performance on the remote sensing datasets. In the Washington DC Mall dataset, CNN-FUS outperforms BSDMF and NMF-DPR in most noisy cases when $s < 8$. Although this method also adopts DPR, it shows different fusion performances on different datasets under different scales compared with the proposed fusion method, which demonstrates that the PMF model adopted by the proposed DPMF method works as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

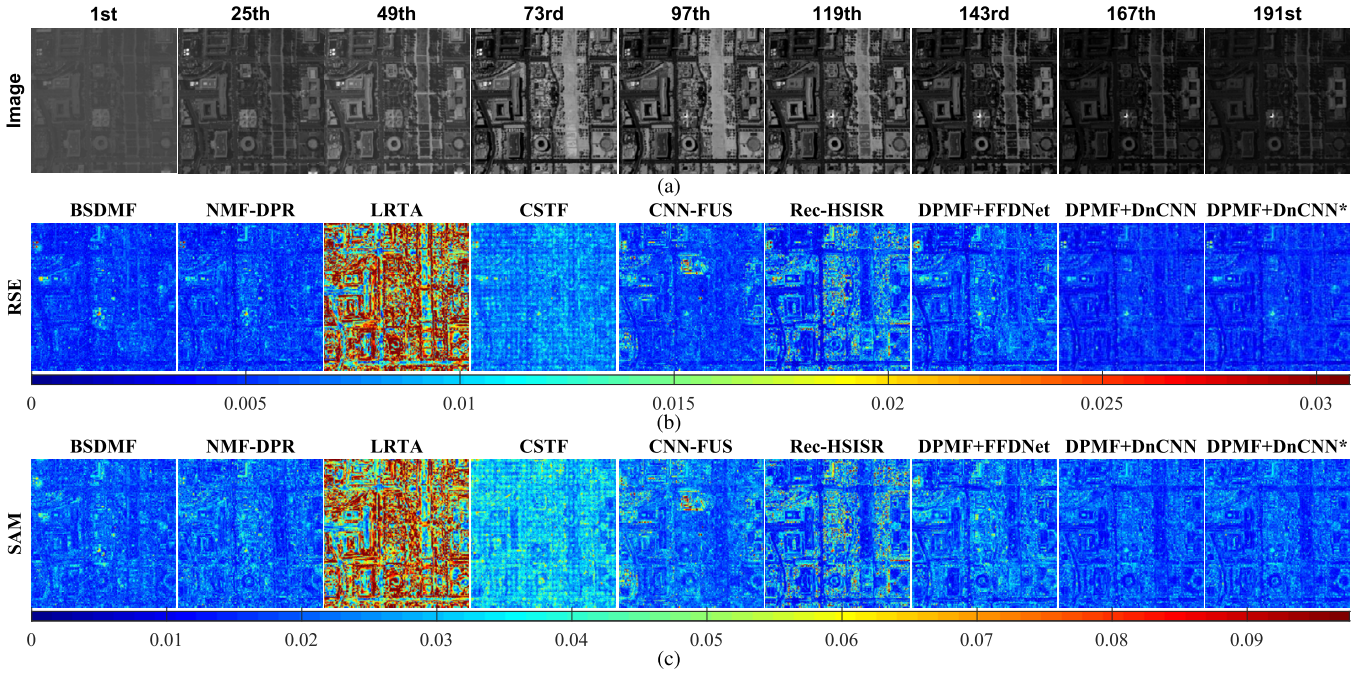


Fig. 10. Fusion results of Washington DC when $s = 8$ ($\text{SNR}_m = 35$ dB and $\text{SNR}_h = 30$ dB). (a) HSI. (b) Root square error. (c) Spectral angle maps.

a key point to largely improve generalization ability and obtain stable fusion performance. For Rec-HSISR, this method shows relatively bad fusion performance in most cases. Besides, we find that the PSNR values of Rec-HSISR first increase and then decrease with the increase in training epochs, i.e., overfitting sometimes occurs in the training process. The main reason for this phenomenon is that Rec-HSISR is designed based on the deep image prior (DIP) network, and it has been reported that the main issue of DIP is overfitting [41].

V. Conclusion

This article proposes a deep HSI-MSI fusion method based on PMF, which contains two highly interpretable iterative steps: the spectral matrix updating step based on least-squares estimation and the abundance matrix updating step based on a 2-D image CNN-based Gaussian denoiser. Thanks to the iPALM algorithm, the proposed method can reach convergence within a few iterations and quickly obtain excellent fusion results. To enhance the generalization ability, we provide solutions for selections of hyperparameters, CNN-based Gaussian denoiser architecture, and training strategy. Experiments show that the proposed fusion method plugged with the DnCNN Gaussian denoiser outperforms other state-of-the-art methods on four different datasets with only a single training process based on 2-D gray images.

References

[1] C. Yu, J. Huang, M. Song, Y. Wang, and C.-I. Chang, "Edge-inferring graph neural network with dynamic task-guided self-diagnosis for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535613.

[2] S. Lee, J.-M. Namgoong, Y. Kim, J. Cha, and J. K. Kim, "Multimodal imaging of laser speckle contrast imaging combined with mosaic filter-based hyperspectral imaging for precise surgical guidance," *IEEE Trans. Biomed. Eng.*, vol. 69, no. 1, pp. 443–452, Jan. 2022.

[3] W. Liu and J. Lee, "An efficient residual learning neural network for hyperspectral image superresolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 4, pp. 1240–1253, Apr. 2019.

[4] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Nonlocal patch tensor sparse representation for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 3034–3047, Jun. 2019.

[5] B. Lin, X. Tao, M. Xu, L. Dong, and J. Lu, "Bayesian hyperspectral and multispectral image fusions via double matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5666–5678, Oct. 2017.

[6] M. Vella, B. Zhang, W. Chen, and J. F. C. Mota, "Enhanced hyperspectral image super-resolution via RGB fusion and TV-TV minimization," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 3837–3841.

[7] Y. Xing, Y. Zhang, S. Yang, and Y. Zhang, "Hyperspectral and multispectral image fusion via variational tensor subspace decomposition," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[8] S. Li, R. Dian, L. Fang, and J. M. Bioucas-Dias, "Fusing hyperspectral and multispectral images via coupled sparse tensor factorization," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 4118–4130, Aug. 2018.

[9] X. Li, Y. Yuan, and Q. Wang, "Hyperspectral and multispectral image fusion via nonlocal low-rank tensor approximation and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 550–562, Jan. 2021.

[10] N. Liu, L. Li, W. Li, R. Tao, J. E. Fowler, and J. Chanussot, "Hyperspectral restoration and fusion with multispectral imagery via low-rank tensor-approximation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7817–7830, Sep. 2021.

[11] T. Gelvez-Barrera, H. Arguello, and A. Foi, "Joint nonlocal, spectral, and similarity low-rank priors for hyperspectral–multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537112.

[12] B. Lin, Y. Zhang, Z. Lin, X. Wang, and H. Huang, "Hyperspectral and multispectral image fusion via nonnegative matrix factorization and deep prior regularization," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2022, pp. 1440–1443.

[13] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.

[14] B. Arad and O. Ben-Shahar, "Sparse recovery of hyperspectral signal from natural RGB images," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2016, pp. 19–34.

[15] J. Qu, Y. Shi, W. Xie, Y. Li, X. Wu, and Q. Du, "MSSL: Hyperspectral and panchromatic images fusion via multiresolution spatial–spectral feature learning networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5504113.

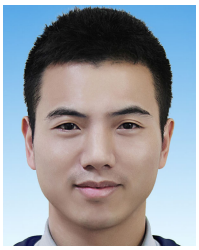
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.

[16] W. Xie, Y. Cui, Y. Li, J. Lei, Q. Du, and J. Li, "HPGAN: Hyperspectral pansharpening using 3-D generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 463–477, Jan. 2021.
[17] W. Dong, Y. Yang, J. Qu, W. Xie, and Y. Li, "Fusion of hyperspectral and panchromatic images using generative adversarial network and image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5508413.
[18] W. Wang, W. Zeng, Y. Huang, X. Ding, and J. Paisley, "Deep blind hyperspectral image fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, vol. 2019, pp. 4149–4158.
[19] Y. Qu, H. Qi, and C. Kwan, "Unsupervised sparse Dirichlet-net for hyperspectral image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2511–2520.
[20] L. Zhang, J. Nie, W. Wei, Y. Zhang, S. Liao, and L. Shao, "Unsupervised adaptation learning for hyperspectral imagery super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3070–3079.
[21] Z. Wang, B. Chen, R. Lu, H. Zhang, H. Liu, and P. K. Varshney, "FusionNet: An unsupervised convolutional variational network for hyperspectral and multispectral image fusion," *IEEE Trans. Image Process.*, vol. 29, pp. 7565–7577, 2020.
[22] R. Dian, S. Li, and X. Kang, "Regularizing hyperspectral and multispectral image fusion by CNN denoiser," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 3, pp. 1124–1135, Mar. 2021.
[23] Z. Wang, B. Chen, H. Zhang, and H. Liu, "Unsupervised hyperspectral and multispectral images fusion based on nonlinear variational probabilistic generative model," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 721–735, Feb. 2022.
[24] J. Liu, Z. Wu, L. Xiao, and X.-J. Wu, "Model inspired autoencoder for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522412.
[25] J. Yang, L. Xiao, Y.-Q. Zhao, and J. C.-W. Chan, "Variational regularization network with attentive deep prior for hyperspectral–multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5508817.
[26] Y. Xu and W. Yin, "A block coordinate descent method for regularized multiconvex optimization with applications to nonnegative tensor factorization and completion," *SIAM J. Imag. Sci.*, vol. 6, no. 3, pp. 1758–1789, 2013.
[27] T. Pock and S. Sabach, "Inertial proximal alternating linearized minimization (iPALM) for nonconvex and nonsmooth problems," *SIAM J. Imag. Sci.*, vol. 9, no. 4, pp. 1756–1787, 2016.
[28] N. Keshava and J. F. Mustard, "Spectral unmixing," *IEEE signal Process. Mag.*, vol. 19, no. 1, pp. 44–57, Aug. 2002.
[29] C. Lanaras, E. Baltsavias, and K. Schindler, "Hyperspectral super-resolution by coupled spectral unmixing," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 3586–3594.
[30] C.-H. Lin, F. Ma, C.-Y. Chi, and C.-H. Hsieh, "A convex optimization-based coupled nonnegative matrix factorization algorithm for hyperspectral and multispectral data fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1652–1667, Mar. 2018.
[31] W. Dong et al., "Hyperspectral image super-resolution via non-negative structured sparse representation," *IEEE Trans. Image Process.*, vol. 25, no. 5, pp. 2337–2352, May 2016.
[32] A. Mnih and R. Salakhutdinov, "Probabilistic matrix factorization," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2007, pp. 1257–1264.
[33] J. M. P. Nascimento and J. M. Bioucas-Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.
[34] B. Lin, X. Tao, and J. Lu, "Hyperspectral image denoising via matrix factorization and deep prior regularization," *IEEE Trans. Image Process.*, vol. 29, pp. 565–578, 2020.
[35] K. Zhang, W. Zuo, S. Gu, and L. Zhang, "Learning deep CNN denoiser prior for image restoration," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2808–2817.
[36] G. H. Golub and C. F. Van Loan, *Matrix Computations*, vol. 3. Baltimore, MD, USA: Johns Hopkins Univ. Press, 2012.
[37] W. Dong, P. Wang, W. Yin, and G. Shi, "Denoising prior driven deep neural network for image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 10, pp. 2305–2318, Oct. 2018.
[38] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.
[39] K. Zhang, W. Zuo, and L. Zhang, "FFDNet: Toward a fast and flexible solution for CNN-based image denoising," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4608–4622, Sep. 2018.
[40] W. Wei, J. Nie, L. Zhang, and Y. Zhang, "Unsupervised recurrent hyperspectral imagery super-resolution using pixel-aware refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500315.
[41] H. V. Nguyen, M. O. Ulfarsson, J. R. Sveinsson, and M. Dalla Mura, "Deep SURE for unsupervised remote sensing image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5412613.



Baihong Lin received the B.S. degree from the Beijing University of Aeronautics and Astronautics, Beijing, China, in 2013, and the Ph.D. degree from Tsinghua University, Beijing, in 2018.

He is currently an Assistant Professor with Sun Yat-sen University, Shenzhen, China. His major research interests include image processing, computer vision, and machine learning.



Yulan Guo (Senior Member, IEEE) received the B.E. and Ph.D. degrees from the National University of Defense Technology (NUDT), Changsha, China, in 2008 and 2015, respectively.

He has authored over 100 articles at highly referred journals and conferences. His research interests lie in 3-D vision, low-level vision, and machine learning.

Dr. Guo is a Senior Member of Association for Computing Machinery (ACM). He has served as the Area Chair of IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2023/2021, IEEE International Conference on Computer Vision (ICCV) 2021, and ACM Multimedia 2021. He has served as an Associate Editor for IEEE TRANSACTIONS ON IMAGE PROCESSING, *IET Computer Vision*, *IET Image Processing*, and *Computers & Graphics*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:12:47 UTC from IEEE Xplore. Restrictions apply.